Gross Anatomy of the Scalp

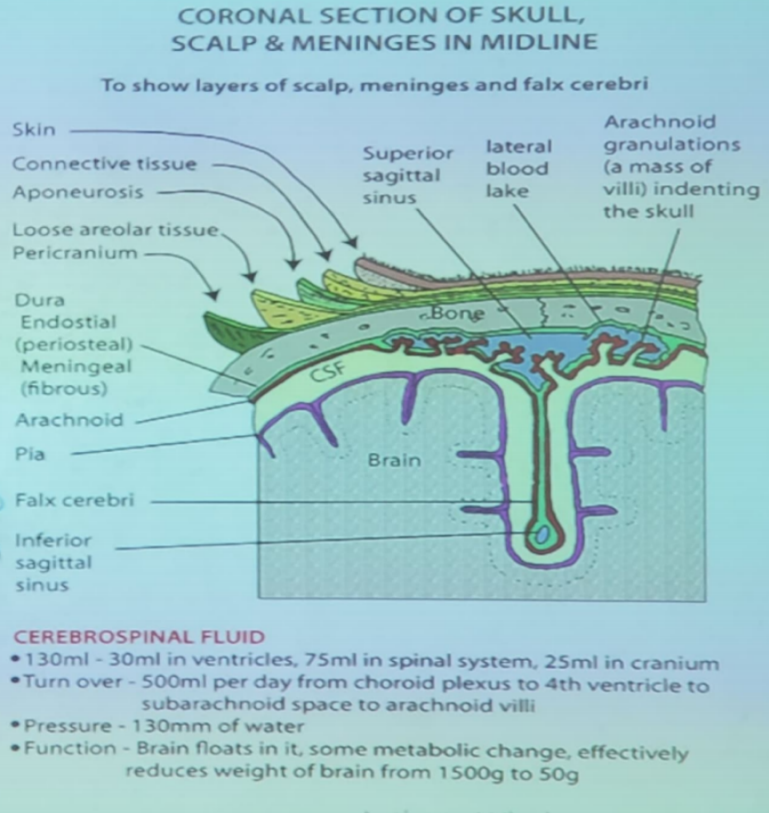
By Dr. Mbaka G.O.

- Head: It is the most cranial part of the body. It contains a structure that coordinate and controls the other parts of the body. The head components is composed of a series of compartments formed by bone and soft tissues. • They are: 1. the cranial cavity, 2. two ears, 3. two orbits , 4. two nasal cavities, and 5. an oral cavity.
- Neck: It is the bridge between the head and the rest of the body. Located between the mandible and the clavicle, connecting the head directly to the torso. It contains proximal esophagus, trachea, thyroid gland, and the parathyroid glands. It provides conduits for blood flow to the brain and head. It transmits nervous signals from the brain to the rest of the body

Scalp: It is the soft tissue structure covering the skull. The anterior limit; eyebrows, posterior limit; superior nuchal line. Lateral limit, superior temporal line.

It is made up of five layers; S- skin, C-connective tissue (superficial fascia containing vessels), A-aponeurosis of occipito-frontalis muscle, L-loose areolar tissue, P-pericranium.

Skin: Studded with hair follicles, firmly adherent to the epicranial aponeurosis through the dense superficial fascia.

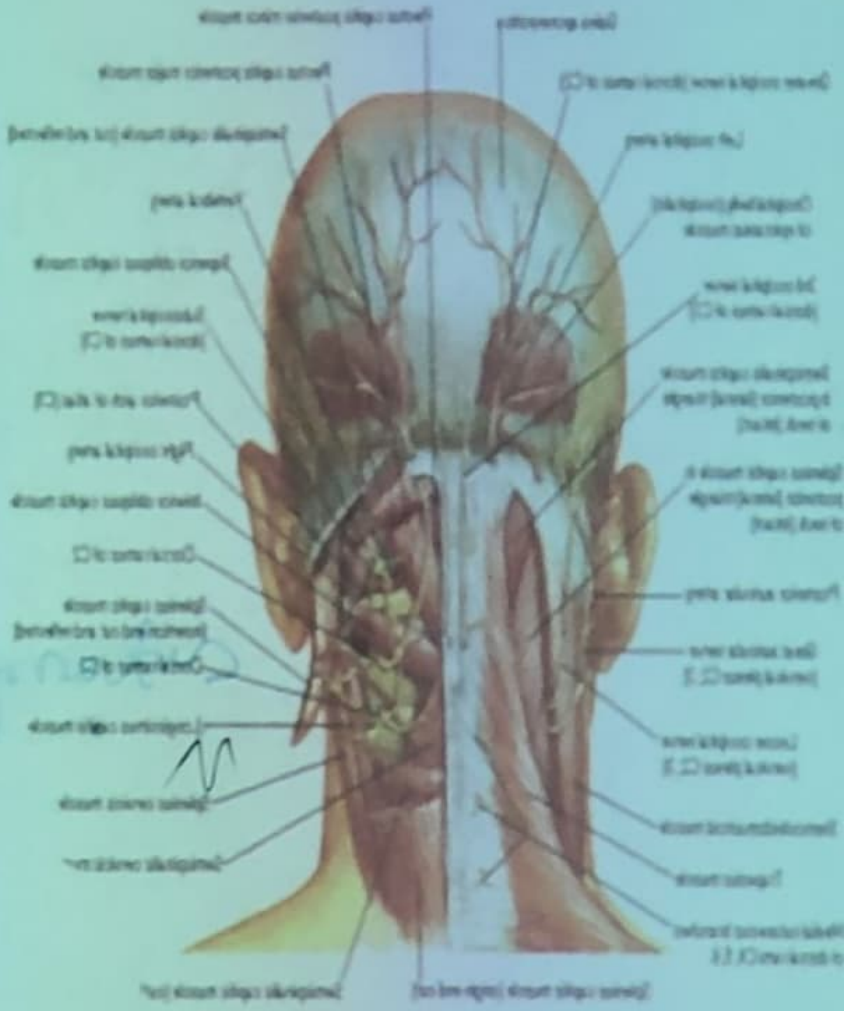


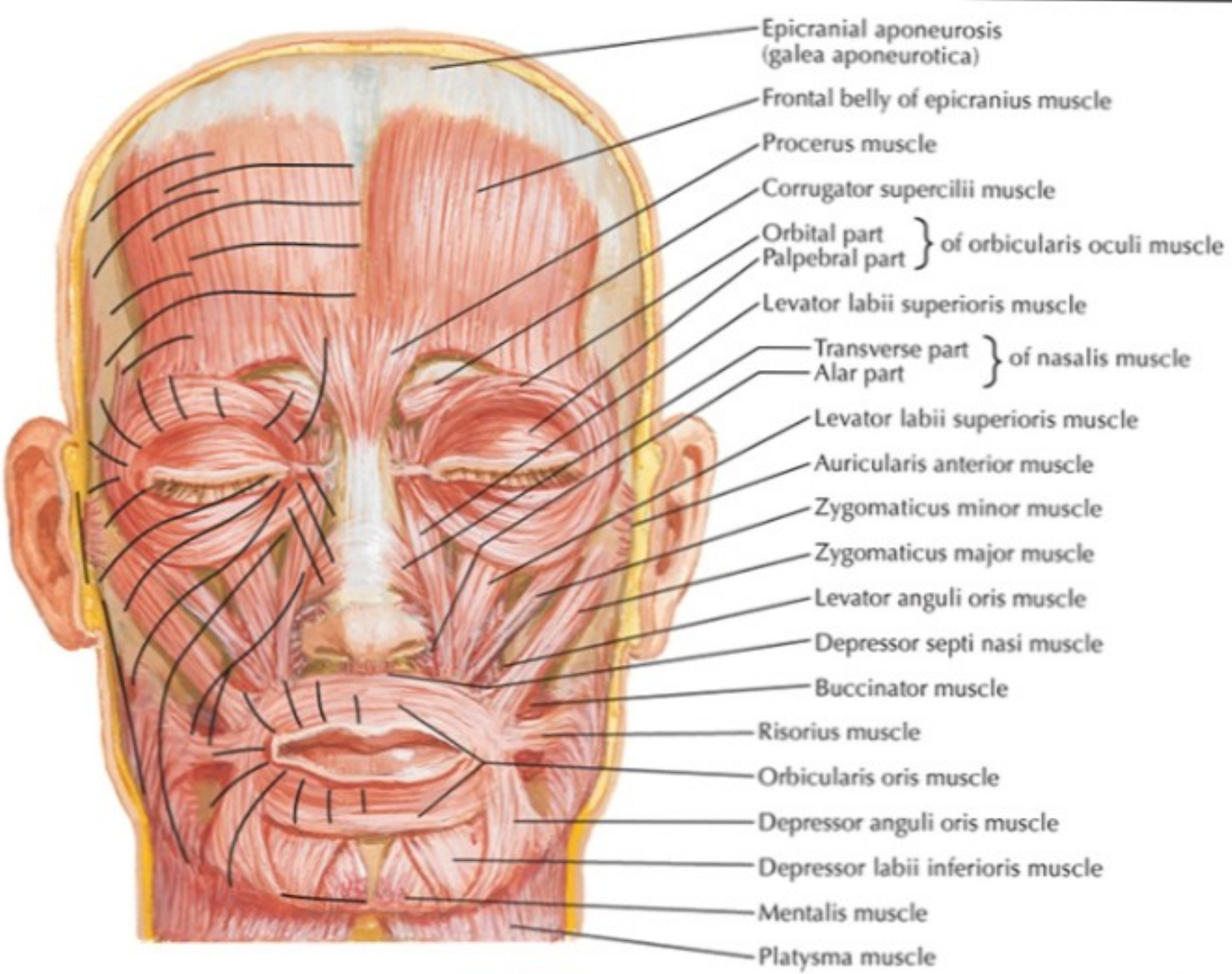
Activate Windows

- Connective tissue (superficial fascia): Tough and fixed to the skin, more fibrous and dense at the centre than periphery of head. It binds the skin to the deep aponeurosis and is rich in nerve and blood supply.
- Aponeurotic layer (*galea aponeurotica*): A thin tough layer covering the vault of the skull and is freely mobile on the pericraneum. Anteriorly meets the frontal belly and posteriorly meets occipital belly.

The occipital bellies are small and separated from each other with each arising from lateral 2/3 of superior nuchal line and supplied by posterior auricular branch of facial nerve.

Frontal bellies are longer, wider and partially united in the median plain. It arises from the skin of forehead mingling with the orbicularis occuli and corrugator supercilli. It is innervated by temporal branch of facial nerve. It wrinkles the skin of forehead.

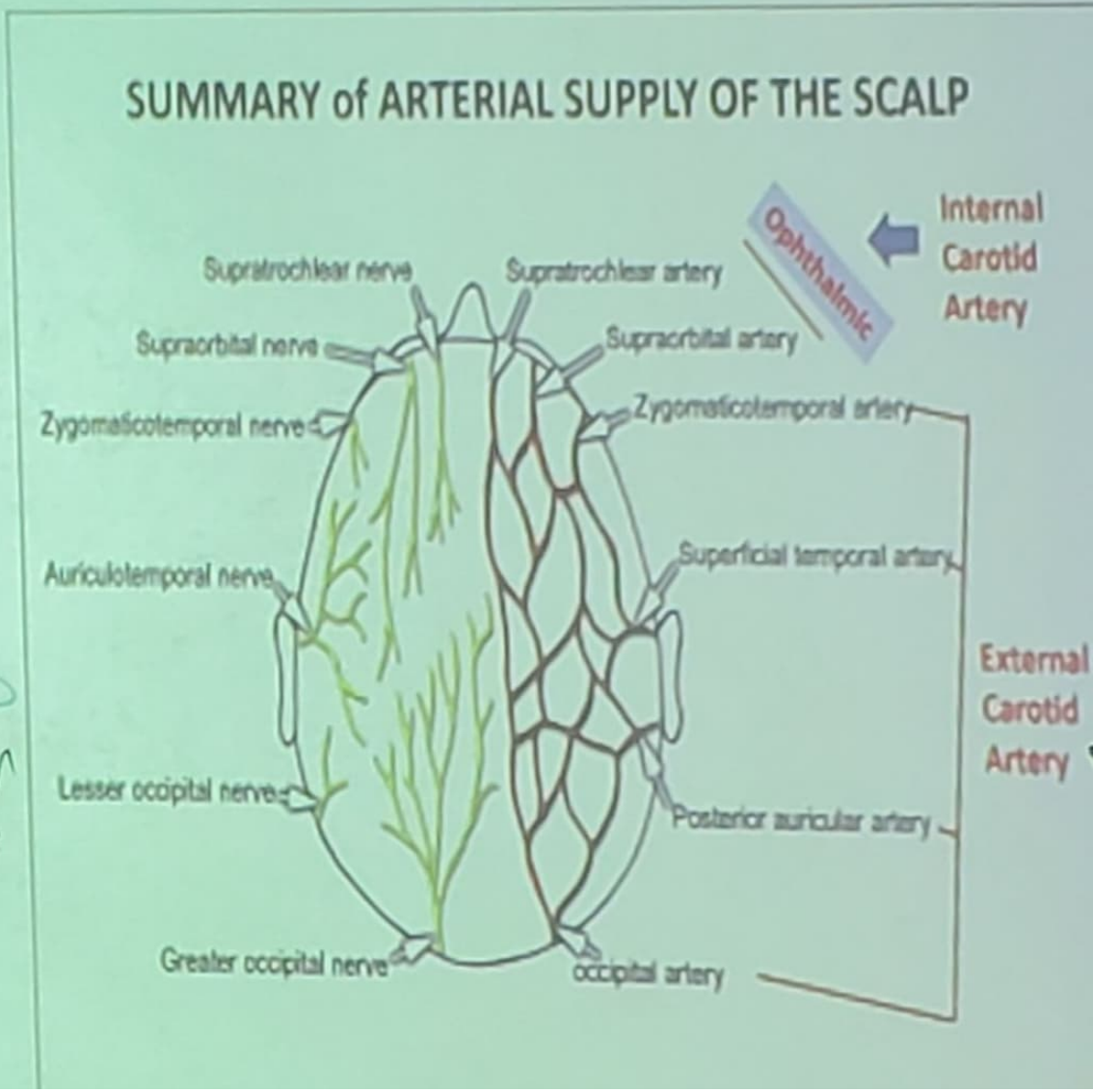



Epicranial aponeurosis
(galea aponeurotica)
Frontal belly of epicranius muscle
Procerus muscle
Corrugator supercilii muscle
Orbital part
Palpebral part
of orbicularis oculi muscle
Levator labii superioris muscle
Transverse part
Alar part
of nasalis muscle
Levator labii superioris muscle
Auricularis anterior muscle
Zygomaticus minor muscle
Zygomaticus major muscle
Levator anguli oris muscle
Depressor septi nasi muscle
Buccinator muscle
Risorius muscle
Orbicularis oris muscle
Depressor anguli oris muscle
Depressor labii inferioris muscle
Mentalis muscle
Platysma muscle

- Loose areolar connective tissue: The layer extends anteriorly to upper eyelid because frontalis has no bony attachment, posteriorly it limits at the highest and posterior nuchal lines and laterally limits at superior temporal lines.

- Pericranium: It is loosely attached to the surface of the bones but with firm adherence to their sutures where the ligament of the suture bind the pericranium to the endocranium.

- Arterial supply: The scalp has rich blood supply. There are five arteries of which three lie in front of the auricle and two behind it. In front of the auricle are supratrochlear, supraorbital and superficial temporal arteries.
- The first two are branches of ophthalmic artery which in turn is a branch of internal carotid artery. The third is the superficial temporal, the terminal branch of external carotid artery.



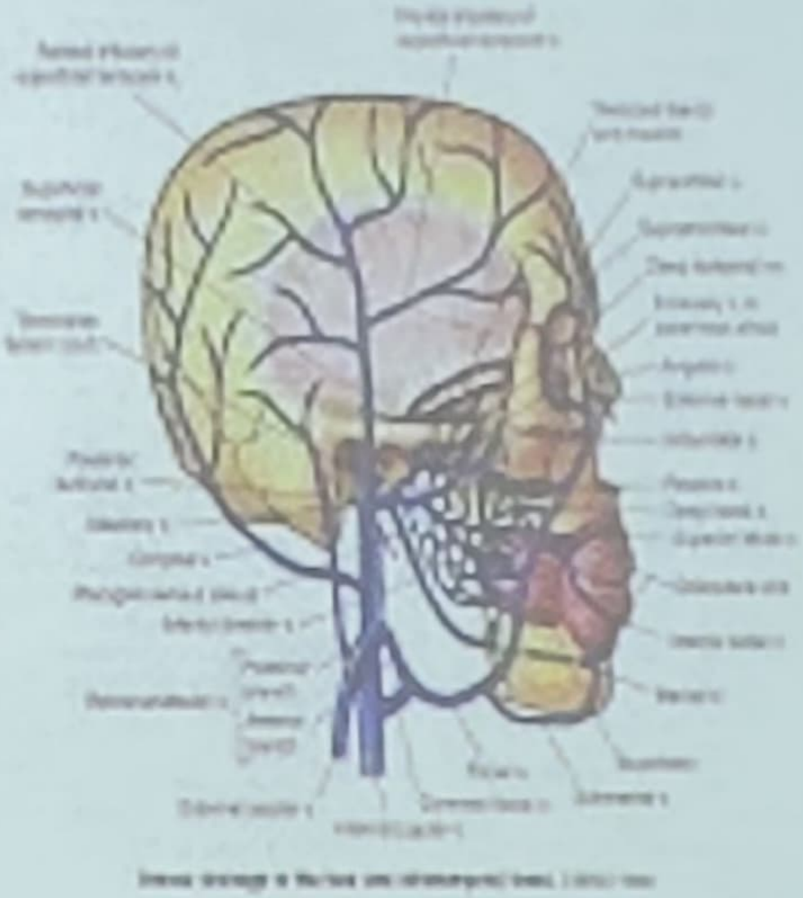
Behind the auricle the scalp is supplied by posterior auricular and occipital arteries both of which are branches of external carotid arteries. The two systems of arterial supply of the scalp anastomosis at the temple.

Venous Drainage: Venous of the scalp accompany their arterial counterparts. The supratrochlear and supraorbital veins units at the medial angle of eye forming the angular vein which continues down as facial vein. The superficial temporal vein descend in front of the tragus to enter parotid gland and in it joins the maxillary vein to form the retromandibular vein. The anterior division of retromandibular vein unites with facial vein to form the common facial vein that drains into the internal jugular vein. The posterior division of retromandibular vein unites with posterior auricular vein to form the external jugular vein and then to subclavian vein.

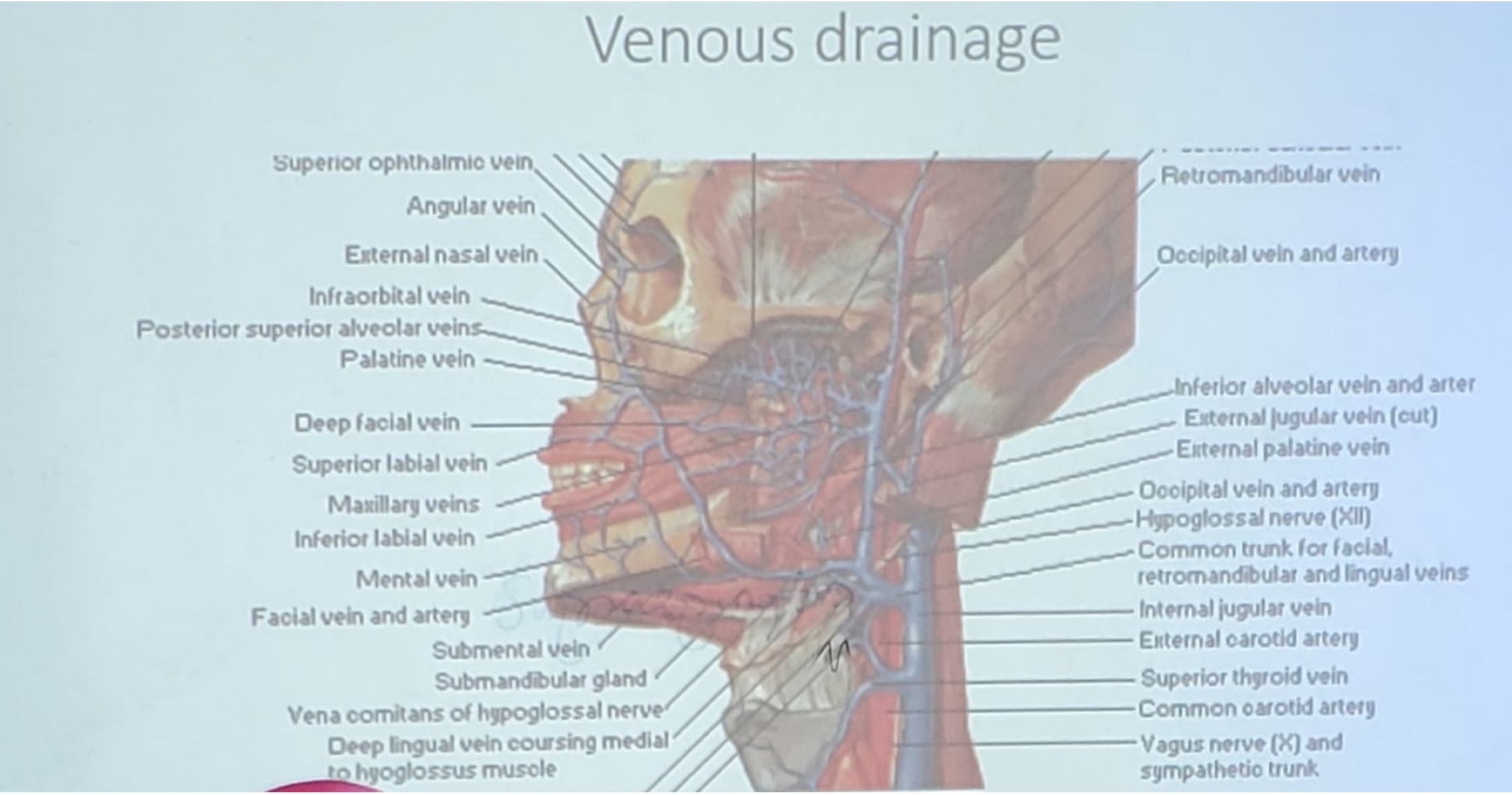
Venous Drainage of the Scalp

- Veins draining the scalp follow a pattern similar to the arteries.

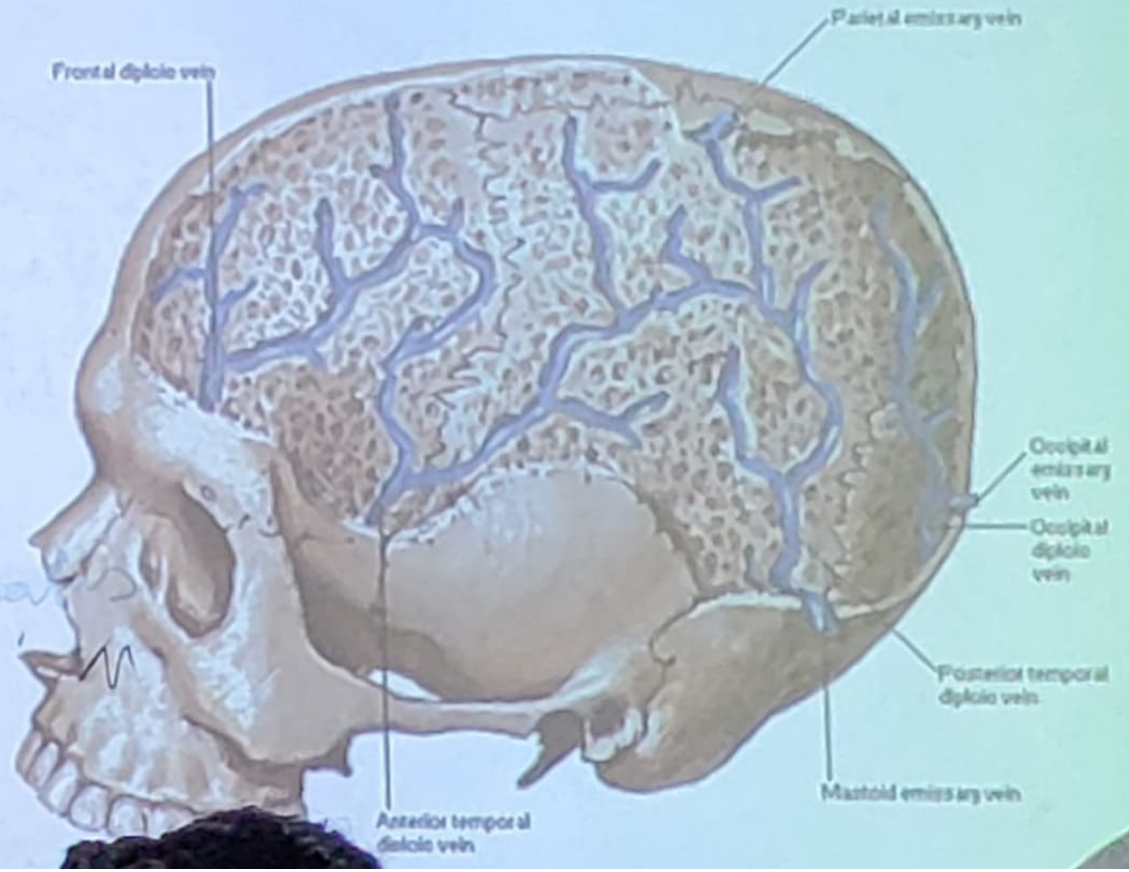
- Of the deep parts of the scalp
  - Via emissary veins that communicates with the dural sinuses.



Venous drainage

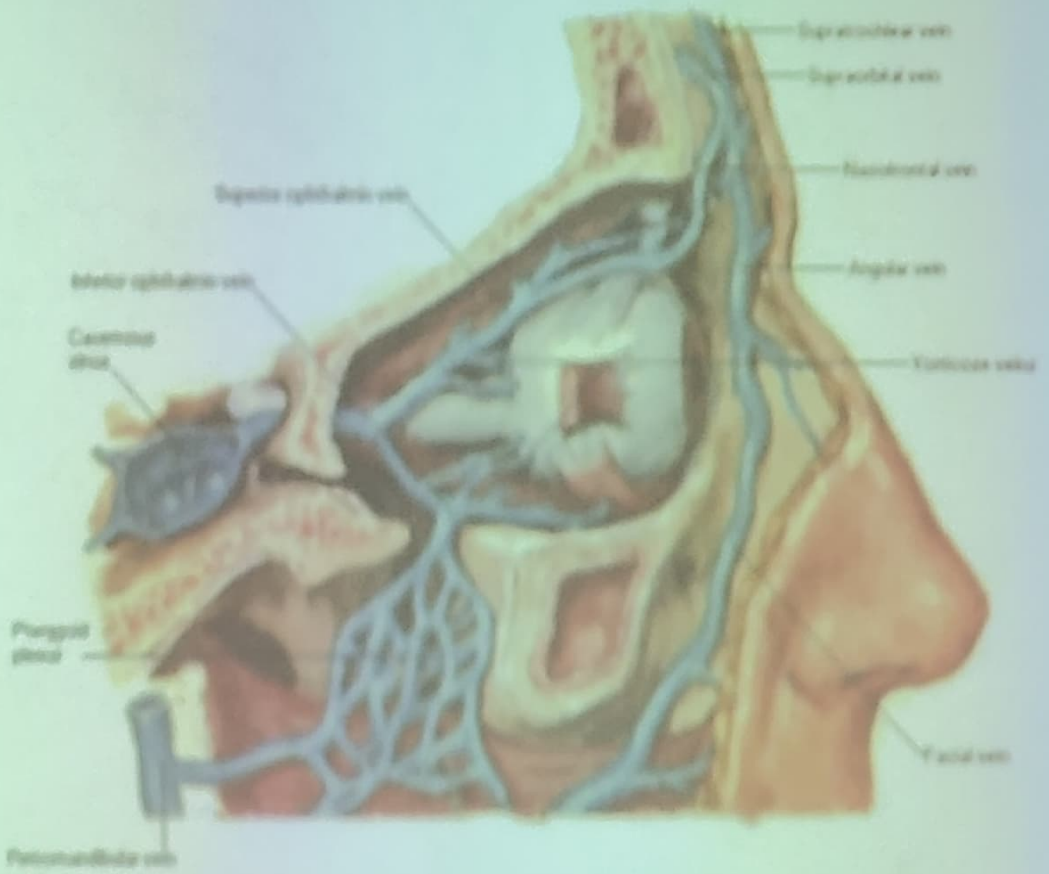


There are also emissary veins which connects intracranial with extracranial veins to equalize pressure. These are parietal emissary vein passing through parietal foramen to superior sagittal sinus and mastoid emissary vein passing through the mastoid foramen to sigmoid sinus.



Danger area of face: Anterior facial vein has no valve, it joins the pterygoid venous plexus through the deep facial vein. The pterygoid venous plexus is connected to the cavernous sinus.

Infection from the upper lip and alae of nose may reach the cavernous sinus and cause thrombosis at the sinus.



Innervations: Scalp is supplied by ten nerves on each side. Five are in front of the auricle and all sensory but one motor while five is at the back of auricle all sensory but one motor.

In front of the auricle- Supratrochlear, ophthalmic div., supraorbital ophthalmic div, zygomatico-temporal branch of maxillary, auriculo-temporal branch of mandibular and temporal branch of facial (motor).

At the back of auricle, Posterior division of great auricular nerve (C2-C3), Lesser occipital (C3), greater occipital nerve C2 (dorsal branch) third occipital (C3 dorsal branch) and posterior auricular branch of facial nerve.

Clinical Anatomy

Cysts: These smooth, firm lumps can develop in your hair follicles. The follicles anchor each hair to your scalp. Cysts grow slowly and usually don't hurt, but sometimes they can get swollen and painful and may ooze a thick fluid with a bad smell.

Ringworm: Is a fungal infection that causes a ring-shaped mark on the skin. It can affect any part of the body, including the scalp.

Ringworm that affects the scalp is known as *tinea capitis*. It can cause a scaly, red, bald patch anywhere on the scalp. This can spread across the scalp, producing many separate spots. Ringworm on the scalp is more likely to affect children than adults.

Folliculitis: The hair on the body and the scalp grow out of hair follicles. Bacteria can enter the skin through damaged hair follicles, causing an infection called *folliculitis*.

If bacteria, a virus, or fungus gets into a hair follicle, you can get this kind of infection. It may look and feel like a pimple, a painful red bump with or without a white head. These sores can be crusty and scabby, too.

People can get folliculitis on their scalp from:

- Shaving or plucking hair on the scalp.
- Frequently touching the scalp.
- Wearing tight hats or other headgear.
- Having hot, damp skin for an extended time.

Activate Windows